C++ Solutions — Iteration and Loop Problems

1) Reverse & Palindrome Check

```
#include <iostream>
using namespace std;
int main(){
    long long n, temp, rev=0;
    cin>>n;
    temp=n;
    while(temp>0){
        rev = rev*10 + temp%10;
        temp/=10;
    }
    if(rev==n) cout<<"Palindrome Number";
    else cout<<"Not Palindrome Number";
}
```

2) Armstrong Number Check

```
#include <iostream>
#include <cmath>
using namespace std;
int main(){
    long long n; cin>>n;
    long long temp=n, sum=0;
    int digits = to_string(n).size();
    while(temp>0){
        int d=temp%10;
        sum += pow(d, digits);
        temp/=10;
    }
    cout<<(sum==n ? "Yes" : "Not");
}
```

3) Fibonacci up to N

```
#include <iostream>
using namespace std;
int main(){
    long long N; cin>>N;
    long long a=0,b=1;
    if(N>=0) cout<<a;
    if(N>=1) cout<<" "<<b;
    long long c=a+b;
    while(c<=N){
        cout<<" "<<c;
        a=b; b=c; c=a+b;
    }
```

```
}
```

4) Skip Multiples of 5 & 7

```
#include <iostream>
using namespace std;
int main(){
    int N; cin>>N;
    for(int i=1;i<=N;i++){
        if(i%5==0 || i%7==0) continue;
        cout<<i;
        if(i<N) cout<<" ";
    }
}
```

5) Sum Digits Until Single Digit

```
#include <iostream>
using namespace std;
int main(){
    long long n; cin>>n;
    while(n>=10){
        long long s=0;
        while(n>0){ s+=n%10; n/=10; }
        n=s;
    }
    cout<<n;
}
```

6) Prime Numbers up to N

```
#include <iostream>
using namespace std;
bool isPrime(int x){
    if(x<2) return false;
    for(int i=2;i*i<=x;i++)
        if(x%i==0) return false;
    return true;
}
int main(){
    int N; cin>>N;
    bool first=true;
    for(int i=2;i<=N;i++){
        if(isPrime(i)){
            if(!first) cout<<" ";
            cout<<i;
            first=false;
        }
    }
}
```